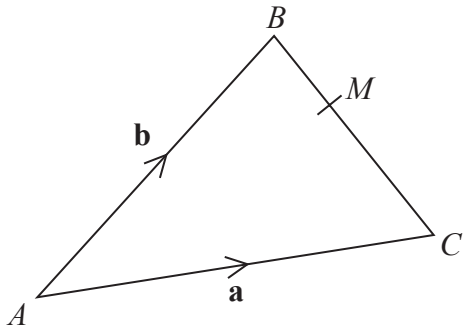
16



M is the point such that $BM : MC$ is $1 : 2$

Here is Charlie's method to find $\overrightarrow{BM}$ in terms of **a** and **b**.

$$\begin{aligned}\overrightarrow{BC} &= \overrightarrow{BA} + \overrightarrow{AC}\\ &= -\mathbf{b} + \mathbf{a}\\ &= \mathbf{a} - \mathbf{b}\\ \overrightarrow{BM} &= \frac{1}{2}\overrightarrow{BC}\\ &= \frac{1}{2}(\mathbf{a} - \mathbf{b})\end{aligned}$$

(a) Evaluate Charlie's method.

BC is correct, but BM is 1 of the 3 equal parts.

Use $\frac{1}{3}(a - b)$, not ½(a – b).

1 mark
Use 1/(1 + 2)

(1)

Martin expands $(2x + 1)(2x - 3)(3x + 2)$

He gets $12x^3 - 4x^2 - 17x + 6$

(b) Explain why Martin's solution cannot be correct.

The constant term must be 1 × (–3) × 2 = –6.

Martin has +6, so his expansion cannot be correct.

1 mark
Constant –6, not +6

(1)

(Total for Question 16 is 2 marks)